Lecture 13
CT System Analysis and Design using Laplace

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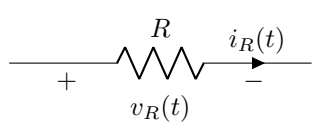
Recall from ECE 2714 that we can analyze electrical circuits that form LTI systems by

- Deriving a governing equation (LCDE)
- Finding the impulse response $h(t)$
- Using convolution to determine the output $y(t)$ given an input $x(t)$

Or alternatively using Fourier analysis

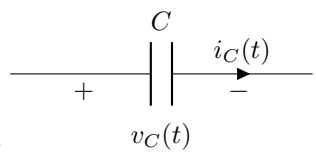
- Determine stability and derive the frequency response $H(j\omega)$
- use convolution theorem to find output $Y(j\omega)$ and use the inverse Fourier transform to find $y(t)$.

Now that we have Laplace as a tool we can simplify this process further. The idea is to go back to circuits and take the Laplace transform of each element's voltage-current model

- Resistor 

$$v_R(t) = R\, i_R(t)$$
$$V_R(s) = R\, I_R(s)$$

Impedance R, Admittance $\frac{1}{R}$

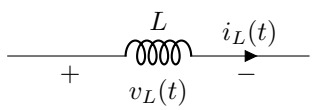
- Capacitor 

$$i_C(t) = C\,\frac{dv_c}{dt}(t)$$
$$I_C(s) = C\left[sV_C(s) - v_c(0^-)\right]$$
$$I_C(s) = CsV_C(s) - Cv_c(0^-)$$

  or rearranging

$$V_C(s) = \frac{1}{Cs}I_C(s) + \frac{v_C(0^-)}{s}$$

  Impedance $\frac{1}{Cs}$, Admittance Cs

- Inductor 

$$v_L(t) = L\,\frac{di_L}{dt}(t)$$
$$V_L(s) = L\left[sI_L(s) - i_L(0^-)\right]$$
$$V_L(s) = LsI_L(s) - Li_L(0^-)$$

  or rearranging

$$I_L(s) = \frac{1}{Ls}V_L(s) + \frac{i_L(0^-)}{s}$$

  Impedance Ls, Admittance $\frac{1}{Ls}$

This gives us new circuit models in Laplace domain with either constant voltage or current sources representing the terms containing auxiliary conditions at $t = 0^-$. Using KVL and KCL with these models gives us a convenient way to go from a circuit directly to a transfer function.

Example: RC with IC

Example: RLC with no I.C.

This works with active (op-amp) circuits as well.

Example: Sallen-key

Most upper level circuit courses will use these techniques. e.g. AC-circuits.

Analyzing Block Diagrams

Recall from ECE 2714 that block diagrams can be used to model systems and implement/realize systems.

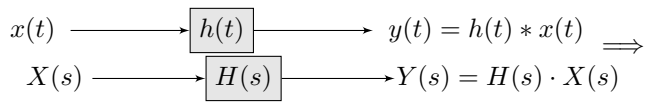


Figure 1: Transformation of input and output to Laplace domain (transfer function) for a single block.

- series connection, the overall transfer function $H(s) = H_1(s) \cdot H_2(s)$

Figure 2: Transformation of input and output to Laplace domain for a series connection.

- parallel connection, the overall transfer function $H(s) = H_1(s) + H_2(s)$

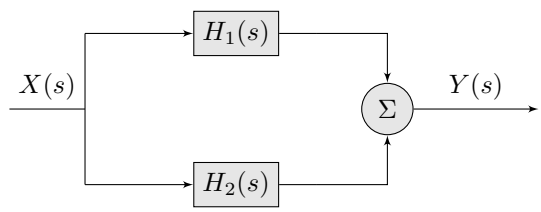


Figure 3: Transformation of input and output to Laplace domain for a parallel connection.

- feedback connection, the overall transfer function $H(s) = \frac{H_1(s)}{1+H_1(s)\cdot H_2(s)}$

We can use block diagrams to derive an overall transfer function from sub-components.

Example: PID controller

Example: DC Motor

System Realization

We can also use block diagrams in the opposite way, to implement, or realize, a given transfer function (for example from a filter design) using three basic building blocks: summation, amplifier (gain), and integrator.

The realization is not unique, with different canonical forms, each with advantages and disadvantages (e.g. reduced number of components and reduced sensitivity to component variation).

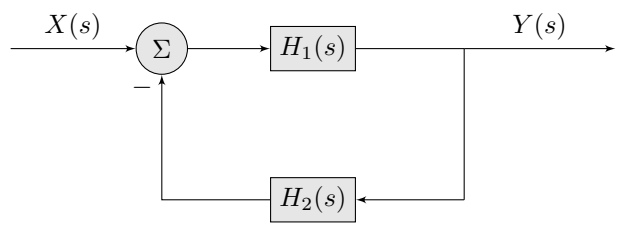


Figure 4: Transformation of input and output to Laplace domain for a feedback connection.

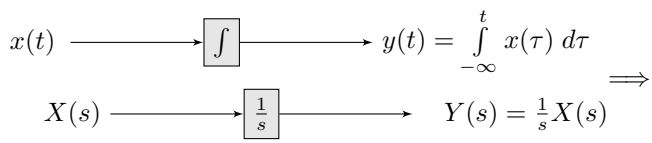


Figure 5: Transformation of an integrator to Laplace domain (transfer function).

Example

Given a transfer function

$$H(s) = \frac{s+a}{s^2+bs+c}$$

for $a, b, c \in \mathbb{R}$, implement the system in terms of summations, amplifiers, and integrators.

TODO: DFI, DFII, transposed DFII, parallel and cascade of second order systems